

2.7

$$(1) F = ABC + \bar{A} + \bar{B} + \bar{C}$$

$$= ABC + \bar{A}(B + \bar{B})(C + \bar{C}) + \bar{B}(A + \bar{A})(C + \bar{C}) + \bar{C}(A + \bar{A})(B + \bar{B})$$

$$= ABC + \bar{A}BC + \bar{A}B\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} + A\bar{B}C + A\bar{B}\bar{C} \\ + A\bar{B}\bar{C} + A\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$= m_7 + m_3 + m_2 + m_1 + m_0 + m_5 + m_4 + \underline{m_1} + \underline{m_0} +$$

$$m_6 + \underline{m_4} + \underline{m_2} + \underline{m_0}$$

$$= \sum m^3 (0, 1, 2, 3, 4, 5, 6, 7)$$

法 = $F = ABC + \bar{A}\bar{B}\bar{C} = 1$
包含全部 8 个最小项

$$(2) F = AB + A\bar{B} + \bar{A}B + \bar{C}\bar{D}$$

$$= AB(C + \bar{C})(D + \bar{D}) + A\bar{B}(C + \bar{C})(D + \bar{D}) + \bar{A}B(C + \bar{C})(D + \bar{D}) + \\ (A + \bar{A})(B + \bar{B})\bar{C}\bar{D}$$

$$\begin{aligned}
 &= ABCD + ABC\bar{D} + AB\bar{C}D + AB\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} \\
 &+ \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D} \\
 &+ \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D}
 \end{aligned}$$

$$\begin{aligned}
 \text{法} &= F = A + \bar{A}B + \bar{C}\bar{D} \\
 &= A + B + \bar{C}\bar{D}
 \end{aligned}$$

$$= m_{15} + m_{14} + m_{13} + m_{12} + m_{11} + m_{10} + m_9 + m_8$$

$$+ m_7 + m_6 + m_5 + m_4 + m_0$$

$$A = \sum m^3(8, 9, 10, 11, 12, 13, 14, 15)$$

$$B = \sum m^3(4, 5, 6, 7, 12, 13, 14, 15)$$

$$= \sum m^4(0, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)$$

$$\bar{C}\bar{D} = \sum m^3(0, 4, 8, 12)$$

$$13) F = \overline{\bar{A}(B + \bar{C})}$$

$$= \bar{\bar{A}} + \overline{(B + \bar{C})}$$

$$= A + \bar{B}\bar{C}$$

$$= A + \bar{B}C$$

$$A = \sum m^3(4, 5, 6, 7)$$

$$\bar{B}C = \sum m^3(1, 5)$$

$$F = \sum m^3(1, 4, 5, 6, 7)$$

$$2.11 \quad (1) F = \sum m^3(0, 1, 2, 4, 5, 7)$$

		B			
		A	00	01	11
C	0	1	1		1
	1	1	1	1	

$$F = \bar{B} + \bar{A}\bar{C} + AC$$

12) $F = \sum m^4 (0, 1, 2, 3, 4, 6, 7, 8, 9, 11, 15)$

D \ B		A			
		00	01	11	10
C	00	1	1	0	1
	01	1	0	0	1
	11	1	1	1	1
	10	1	1	0	0

$F = CD + \bar{A}\bar{D} + \bar{B}\bar{C}$

13) $F = \sum m^4 (3, 4, 5, 7, 9, 13, 14, 15)$

D \ B		A			
		00	01	11	10
C	00				1
	01			1	
	11	1		1	1
	10		1	1	1

$F = \bar{A}\bar{B}\bar{C} + \bar{A}CD + ABC + A\bar{C}D$

2.12.

$$11) F = \sum m^4(0, 2, 7, 13, 15)$$

$$\sum d(1, 3, 4, 5, 6, 8, 10)$$

$\begin{array}{c} B \\ \diagdown \\ D/A \end{array}$					
		00	01	11	10
C	00	1	X	0	X
	01	1	X	1	X
	11	X	1	1	1
	10	X		1	X

$$F = \bar{A} + BD$$

$$12) F = \sum m^4(0, 3, 5, 6, 8, 13) \quad \sum d(1, 4, 10)$$

$\begin{array}{c} B \\ \diagdown \\ D/A \end{array}$					
		00	01	11	10
C	00	1	1		X
	01		X		1
	11	1			
	10	X		1	1

$$F = \bar{B}\bar{C}\bar{D} + \bar{A}B\bar{D} + \bar{A}B\bar{D} + B\bar{C}D$$

$$13) F = \sum m^4(0, 2, 3, 5, 7, 8, 10, 11)$$

$$\sum d(14, 15)$$

$\begin{array}{c} B \\ \diagdown \\ D \end{array} \begin{array}{c} A \\ \diagup \\ C \end{array}$		00	01	11	10
		00	01	11	10
00	00	1	1		
01	01	1	1	x	
11	11	1	1	x	1
10	10				1

$$F = \bar{B}\bar{D} + CD + \bar{A}BD$$

2.B

$$(1) F = ABC + \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

$\begin{array}{c} B \\ \diagdown \\ D \end{array} \begin{array}{c} A \\ \diagup \\ C \end{array}$		00	01	11	10
		00	01	11	10
0	00	1	1		1
1	01	1		1	

$$F = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}\bar{C} + ABC$$

$$(2) F = AC + ABC + A\bar{C} + \bar{A}\bar{B}\bar{C} + BC$$

		B				
		A	00	01	11	10
C	0		1	1	1	
	1			1	1	1

$$\bar{F} = A + \bar{B}\bar{C} + BC$$

$$(3) F = \bar{B}\bar{D} + ABCD + \bar{A}\bar{B}\bar{C}$$

		B				
		A	00	01	11	10
D	C					
	00	1	1			
	01	1	1			
	11			1		
	10	1				

$$F = \bar{B}\bar{D} + \bar{A}\bar{B}\bar{C} + ABCD$$